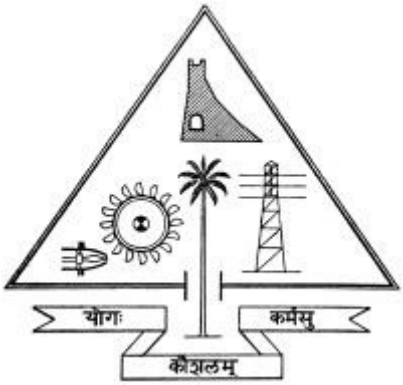
AI-POWERED CODE DOCUMENT GENERATOR

Main Project Report submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications of the APJ Abdul Kalam Technological University

submitted by

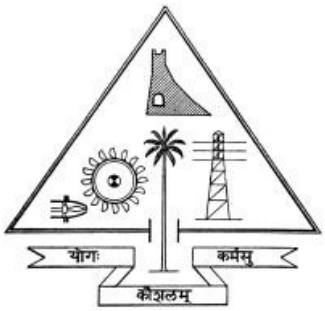
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CERTIFICATE

This is to certify that the main project titled "AI-POWERED CODE DOCUMENT GENERATOR" is a bonafide work done by ARYA K A (TCR24MCA-2020) under my supervision and guidance, and is submitted in March 2026 in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications from APJ Abdul Kalam Technological University (KTU).

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Place : Thrissur
Date : 31-03-2026

DECLARATION

I hereby declare that the main project named, AI-POWERED CODE DOCUMENT GENERATOR, is my own work and that, to the best of my knowledge and belief, it contains no material previously published by another person nor material which has been accepted for the award of any other degree or course of the university or any other institute of higher learning, except where due acknowledgment and reference has been made in the text.

Place : Thrissur
Date : 31-03-2026

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ABSTRACT

Maintaining software project documentation accurately and up to date is hard because code changes often, and writing documentation by hand takes a lot of time. Outdated documentation can make code harder to understand, require more effort to keep up to date, and pose challenges for new developers who want to join a project. To solve these problems, this project shows off an AI-Powered Code Document Generator that looks at the source code of a software project and automatically creates structured documentation.

The system can read code written in different programming languages from local directories or optional version-controlled repositories. It uses Abstract Syntax Tree (AST) parsing to get information about the structure like functions, classes, and dependencies, which help you understand the codebase better. After that, these extracted features are given to a Large Language Model (LLM), which turns the technical details into clear, easy-to-read documentation while keeping the accuracy and context.

The system makes a lot of different documents, such as a README file, architectural descriptions, file-wise explanations, and a list of recent changes. The proposed method makes it easier for developers to maintain software by automating the documentation process without changing the original source code. In general, this project shows how well artificial intelligence can be used to make technical documentation that is accurate, well-organized, and easy to understand.

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LIST OF ABBREVIATIONS

AI	–	Artificial Intelligence
API	–	Application Programming Interface
AST	–	Abstract Syntax Tree
BART	–	Bidirectional and Auto-Regressive Transformers
BLEU	–	Bilingual Evaluation Understudy
CD	–	Continous Integration
CI	–	Continous Deployment
CSS	–	Cascading Style Sheets
DFD	–	Data Flow Diagram
GPT	–	Generative Pre-trained Transformer
HTML	–	HyperText Markup Language
IDE	–	Integrated Development Environment
LLM	–	Large Language Model
METEOR	–	Metric for Evaluation of Translation with Explicit Ordering
Q/A	–	Question and Answer
REST API	–	Representational State Transfer Application Programming Interface
RoBERTa	–	Robustly Optimized BERT Approach
ROUGE	–	Recall-Oriented Understudy for Gisting Evaluation
T5	–	Text-to-Text Transfer Transformer
URL	–	Uniform Resource Locator
UTF-8	–	8-bit Unicode Transformation Format
VS Code	–	Visual Studio Code

CHAPTER 1
INTRODUCTION

The rise in software system's development and the rising complexity in application development processes has made the need for good code documentation necessary. Software systems tend to become more advanced in terms of features, connections and development teams. Code documentation helps make code readable, encourages teamwork between programmers and guarantees software system maintainability in the future. Documentation acts as a link between programmers by offering an explanation of system features, design choices and coding processes. In large projects involving many programmers working on the project for a long time, documentation is vital to ensuring consistency and minimizing misunderstandings. Nonetheless, keeping documentation up to date and accurate is difficult because there are many modifications to code and it is time-consuming to write documentation manually.

Often, in practice, in many software development projects, it is observed that documentation tends to fall behind and there are chances that the documentation might not be complete for the software developed. In such situations, the developers do not have enough time to document the changes made to the codebase, owing to their focus being diverted to implementing features and fixing bugs. It also results in an inconsistency in the actual state of the codebase and what is described in the documentations related to the code. This makes it challenging for developers to understand the behavior of the system and hence spend extra time debugging and analyzing the actual code.

For addressing these difficulties, a set of latest development of Artificial Intelligence (AI) technologies can be used to automate documentation creation. The analytical technique of Abstract Syntax Tree (AST) is one of the most useful ways of analyzing source codes by building an abstract syntactic tree. Such components include functions, classes, variables and control statements, thus helping to have a clear understanding of the structure of source codes. Compared to traditional text approaches, analysis through AST allows to get more precise results and insights on behavior of the source codes analyzed.

Apart from the AST-based analysis, another way of gaining deeper insight into code by analyzing relationships among its different elements is by means of graphical representation. In this method, nodes are represented using objects such as functions, classes and modules, while edges are represented using relationships between them, such as function call dependencies and inheritance. Graphs can be very useful to identify relationships and patterns in the form of dependencies, control flow and the most important components in the system. So, together with the AST analysis method, graphs give a broader perspective on the code base.

In addition, Large Language Models (LLMs) have brought another evolution in the documentation process, where the complex and complicated structure of code can be translated into understandable and readable text. The detailed grasp of the semantics of the code by LLMs is useful in making fine descriptions, summaries and documentation of the code itself. Using structured input based on the analysis of an abstract syntax tree and graphs, LLMs can provide precise, consistent and contextually relevant documentation, README files and architectural documentation of modules and functions.

The goal of this project is to create an intelligent system for automatic generation of documentation based on source code. This will be achieved through the use of AST-based parsing and modeling techniques, as well as natural language models, to reduce human interference, enhance software readability and maintainability. This intelligent system will aid developers in

creating accurate and updated documentation without having to modify the source code.

The increasing demand for intelligent and scalable documentation highlights the limitations of conventional methods in handling complex software systems. Manual documentation processes are often slow, inconsistent and unable to keep up with rapid development cycles and frequent software updates. As a result, maintaining accurate and up-to-date documentation becomes a significant challenge.

In contrast, integrating structural analysis techniques with state-of-the-art language models offers a more effective solution by enabling automated and context-aware documentation generation. Therefore, reviewing the existing literature is essential to identify current approaches, their limitations and potential areas for improvement, which are further explored in the following chapter.

CHAPTER 2
LITERATURE REVIEW

Maintaining accurate and up-to-date software documentation remains one of the greatest challenges in contemporary software development. As software systems continue to evolve through continuous integration and agile methodologies, documentation created manually often becomes inconsistent with the source code. This mismatch between code and its documentation reduces code legibility and makes it challenging for new developers joining a codebase. To address these problems, researchers have recently made efforts toward automating document generation using artificial intelligence. In particular, combining structural program analysis with LLM has emerged as a promising solution. This chapter reviews the most relevant literature, organized around key themes including code representation strategies, repository-level documentation approaches, empirical evaluations of language models, and practical deployment considerations.

A fundamental challenge in automated code documentation is how to represent source code in a way that captures its meaning and structure for machine processing. Early approaches treated code as simple sequences of tokens, but this ignored the rich syntactic structure that distinguishes programming languages from natural language. Researchers have increasingly recognized that AST offer a more powerful representation, as they break code down into meaningful components such as functions, classes, and control statements while preserving hierarchical relationships. However, simply serializing ASTs into sequences for neural network processing loses important positional relationships between nodes, including hierarchical parent-child relations and sibling order. To address this limitation, researchers have

proposed tree positional embeddings that inject both vertical and horizontal positional information for each node in the AST, along with specialized syntax-tree attention mechanisms that focus on syntactically important code tokens. This enhanced AST encoding has demonstrated consistent improvements across multiple programming languages and evaluation metrics, outperforming state-of-the-art models by significant margins on both Java and Python benchmarks [1]. Beyond traditional AST representations, researchers have explored treating code as graphs to capture relationships that tree structures cannot adequately represent. When ASTs are transformed into graph representations where nodes correspond to syntactic units and edges represent hierarchical relationships, graph neural networks can extract structural features through neural message passing with neighborhood aggregation. This graph-based perspective preserves connectivity information that tree representations miss, such as complex dependency relationships and recursive patterns, and has achieved high accuracy in code understanding tasks [2]. Additionally, incorporating code metrics such as complexity measures, API (Application Programming Interface) popularity, and Halstead metrics into documentation models has been shown to improve generation quality, suggesting that structural characteristics beyond raw syntax contribute meaningfully to code comprehension [3].

The integration of multiple complementary code representations has emerged as another important direction for improving documentation quality. Rather than relying on a single representation, combining token sequences with type-augmented ASTs and control flow graphs provides a more complete picture of code semantics. Type augmentation enriches ASTs with type information about variables and expressions, while control flow graphs capture program execution order and branching structure. Hierarchical attention mechanisms have proven effective for integrating these diverse representations, with one attention layer operating at the token level within statements and another layer operating at the statement level within functions. This hierarchical structure mirrors the natural hierarchy of code,

where tokens form statements and statements form functions. Reinforcement learning has also been employed to address the exposure bias problem in sequence generation, where models trained to predict the next word given ground truth context must rely on their own potentially incorrect predictions during actual use. Actor-critic networks jointly optimize word selection and evaluate overall sequence quality, leading to substantial improvements over maximum-likelihood training approaches [4].

Documenting entire software repositories presents challenges beyond those of individual function summarization. Understanding a full software project requires comprehending inter-file dependencies, system architecture, and how different modules collaborate to implement overall functionality. One framework addresses this by constructing a project tree that reflects the repository's directory structure, then extracting reference relationships between code objects to form a directed graph that captures semantic call relationships. These reference relationships serve as natural contextual learning examples, teaching language models how different parts of the codebase interact. The framework generates structured documentation covering functionality, parameters, code descriptions, considerations, and examples, using prompt templates that incorporate project trees, code snippets, reference relationships, and metadata [5]. A feature-oriented approach to repository summarization offers an alternative to directory-based organization. Rather than relying solely on file structure, this method clusters methods and files by both dependency relationships and semantic similarity to identify functional features that naturally span multiple files. The clustering process uses community detection algorithms on graphs where nodes represent code elements and edges represent both structural dependencies and semantic similarities. Once features are identified, large language models generate descriptive documentation for each feature, and traceability links are preserved between features and the code elements that implement them. This feature-oriented organization has been shown to increase the rate of completely covered features in manual documentation and

significantly improve the recall of traceability links between documentation and code [6].

Beyond generating documentation directly from code, researchers have also explored the related task of generating code examples to complement existing documentation. This work recognizes that good documentation includes not only descriptions of what code does but also concrete examples of how to use it. Using language models pre-trained on both natural and programming languages, researchers have generated code examples from source code and documentation inputs. The effectiveness of this approach has been demonstrated through metrics such as passability, measuring whether generated examples execute without error, and relevance, measuring whether examples properly address the target method and documentation. A particularly significant finding is that incorporating compiler error logs into the input improves passability substantially, as the error messages provide feedback that allows the model to correct mistakes in subsequent attempts. This error-feedback mechanism demonstrates the value of iterative refinement based on execution results [7].

Empirical evaluation of different language models for documentation tasks has provided valuable guidance for model selection. Comparative studies have examined models such as GPT-2 (Generative Pre-trained Transformer–version 2) and RoBERTa (Robustly Optimized BERT Pretraining Approach), finding that RoBERTa consistently outperforms GPT-2 in both accuracy and loss metrics. The superior performance of RoBERTa is attributed to its ability to detect semantic connections in code, achieving near-perfect accuracy on documentation generation tasks while maintaining dramatically lower training and validation losses. These results highlight the importance of model architecture choices and the value of fine-tuning with domain-specific datasets for documentation generation [8]. Similar comparative work on T5 (Text-to-Text Transfer Transformer) and BART (Bidirectional and Auto-Regressive Transformer) has shown that it generates more concise and context-aware comments, making it more reliable for capturing the single

intent of code snippets. However, T5 offers advantages in training efficiency, presenting a trade-off between output quality and computational resources. The use of multiple evaluation metrics including BLEU (Bilingual Evaluation Understudy), ROUGE (Recall-Oriented Understudy for Gisting Evaluation), METEOR (Metric for Evaluation of Translation with Explicit ORdering), and Smoothed BLEU provides a comprehensive framework for assessing documentation quality across different dimensions [9].

The practical deployment of automated documentation systems in real-world development workflows has also received attention. One system integrates AST generation from a compiler with a large language model within a continuous integration and continuous delivery pipeline, triggered on every code push to automatically generate updated documentation. This integration with version control systems ensures that documentation remains synchronized with code changes, addressing the documentation debt problem directly. A survey of experienced developers found that concise documentation was strongly preferred over verbose output, with developers rating terseness as more important for comprehensibility than additional detail. This finding has important implications for prompt design and model configuration in automated documentation systems, suggesting that generating shorter, focused documentation may be more valuable than producing comprehensive but lengthy explanations [10].

In summary, the reviewed literature establishes that effective automated code documentation requires careful attention to code representation, with AST-based, graph-based, and metric-enriched approaches each contributing valuable perspectives. Repository-level documentation benefits from both directory-based and feature-oriented organization strategies. Empirical studies provide guidance on model selection, with different models offering trade-offs between output quality and computational efficiency. Practical deployment in CI/CD (Continous Integration/ Continous Deployment) pipelines has demonstrated feasibility, with user preferences favoring concise documentation. The proposed AI-Powered Code Document Generator

builds on these findings by combining Tree-sitter-based AST parsing across fourteen programming languages, NetworkX-based graph analysis for capturing function call relationships, and a locally-hosted LLM to generate comprehensive documentation including README files, architectural descriptions, file-wise explanations, and recent change histories without modifying source code.

Even though all the above-mentioned methods are required to create the proposed system, their implementation needs a certain environment with resources and tools prepared beforehand. That is why the proper setting up of the development environment will be important and this point is described in the next chapter.

CHAPTER 3
ENVIRONMENTAL STUDY

The environmental settings of the project are discussed in this chapter as it is essential in ensuring that the AI-Powered Code Document Generator system is successfully developed, implemented and assessed. Environment of the project is important because it helps system perform its functions effectively while ensuring that all its component are integrated.

Environment entails both hardware and software pre-requisites that may be required by the system to operate successfully under ideal scenarios. It emphasizes all the tools, frameworks and platforms used during system development. The configuration of these elements optimizes the performance of the system.

3.1 System Configuration

System configuration means the collection of hardware and software components necessary for developing and executing the system. The proper configuration helps to ensure that the system works efficiently and effectively. Source code is processed easily with the help of the right configuration. At the same time, it helps ensure good interaction between all modules within the system.

Operations that the system may perform include parsing, AST creation, feature extraction, and communication with LLM for documentation generation. Therefore, system configuration, based on an adequately developed environment, is essential for ensuring better overall system performance and less processing time during the testing and implementation phases so that, eventually, a system can run without glitches and lags.

3.1.1 Hardware Requirements

The hardware requirements outline the minimum resources necessary to operate the system. These resources are crucial for performing code analysis, processing tasks and running machine learning operations.

The key hardware requirements are detailed as follows:

- Processor: Intel® Core™ i5 Processor / AMD Ryzen™ 5 Processor or higher, to efficiently handle code processing tasks.
- Memory: Minimum 8 GB RAM, to ensure smooth execution of analysis and model operations.
- Storage: 50–100 GB of free disk space, for storing code files and generated outputs.

3.1.2 Software Requirements

The requirements for software specify the necessary platforms, languages and libraries that will be used in running the system. The use of these tools, activities like parsing, integration and documentation become easy. Compatibility assurance is relevant to ensure that the various components of the system communicate effectively.

The key software requirements are detailed as follows:

- Operating System: Windows 10, Ubuntu 20.04 LTS or macOS, to support cross-platform execution.
- Programming Language: Python 3.8 or higher, used for implementing the entire system.
- Version Control System: Git (optional), for managing repository-based inputs and commit history.
- Required Python Libraries:
  - Tree-sitter – a parser that converts source code into an AST, enabling precise analysis of functions, classes and other structures.

  - Markdown – a lightweight markup language for generating clear and easily readable documentation from code.
  - NetworkX – a Python library for creating and analyzing graphs, useful for visualizing relationships between functions, classes, and modules in source code.

3.1.3 Software Requirements for Development

Besides the execution specifications, some indispensable are the tools for development, testing and maintenance. They assist in writing and debugging programs, as well as managing the code appropriately. Besides, the tools help in maintaining versions and assembling different systems together.

The key development software requirements for development are detailed as follows:

- Operating System: Windows 10, Ubuntu 20.04 LTS or macOS, ensuring cross-platform compatibility.
- Back End: Python 3.8 or later, serving as the core implementation language.
- IDE Used: Visual Studio Code, used for development and debugging tasks.
- Version Control: Git, for managing source code and repositories.
- Development Tools and Frameworks:
  - Tree-sitter – for AST-based code analysis.
  - Large Language Model (Ollama DeepSeek) – for generating documentation efficiently.

3.2 Software Specification

The following section provides a detailed description of different software components and technologies that are used to design the system under

discussion. This is because each of the software components performs specific tasks that shall be useful when analyzing the code and preparing the documentation. Each software component is made to perform particular operations that are useful in code analysis and documentation preparation and the use of technologies is meant to ensure a smooth integration process.

3.2.1 Python 3.8

The choice of programming language for this project has been Python due to its ease-of-use, flexibility and the availability of community support for artificial intelligence and software analysis utilities. Code parsing, feature extraction, creation of graphs, interfacing with LLMs and the automation of documentation creation is done using Python. The large number of libraries that Python offers, including the NetworkX library for working with graphs and many utility libraries for working with files, helps in implementing difficult functionalities efficiently.

3.2.2 Visual Studio Code

Visual Studio Code (VS Code) has been picked as the primary Integrated Development Environment (IDE) that will be used during development, management and system maintenance. The software offers a very efficient, powerful platform with full Python support and a multitude of development capabilities. These features include syntax highlighting, code suggestions, debugging tools, command prompt integration and many other features that aid in an organized and efficient software development process.

The editor provides a broad set of features including Python functionality, Git functionality and formatting utilities that make coding more efficient. The inbuilt versioning capabilities of the software enable the use of Git projects. In addition, the cross-platform interface provided by VS Code makes it possible to use it across various operating system platforms in developing, debugging and maintaining the proposed system.

3.2.3 Tree-sitter

Tree-sitter is used as an important parser for parsing the code and creating its AST. Tree-sitter offers fast and incremental parsing abilities, making it easy to analyze the code efficiently. Tree-sitter is used to extract many code elements such as file, module, class, function, import and dependency based on the particular programming language. Moreover, Tree-sitter can handle several programming languages, which include Python, Java, C, C++, JavaScript, TypeScript, Go, Rust, Ruby, PHP and Kotlin.

The code tree created via Tree-sitter is highly extensive and provides syntactical as well as semantical elements. Hence, this is beneficial in detecting functions together with arguments and return types with more accuracy. Therefore, the code tree serves as a foundation for further processing of the code by the remaining modules, such as graph generation and feature extraction, more efficiently.

3.2.4 Large Language Model(LLM)

LLM, is utilized to articulate representations of codes into a comprehensive, human-readable format of documentation that interprets and ensures the understanding of the code units correctly. The output is human readable and includes project summaries, architectural summaries and provides a summary for each function.

The LLM then uses its advanced linguistic comprehension to understand programming concepts and code relationships. It has the power to produce human-readable narratives that are brief in length and dependent on the context yet resembling in appearance to human-generated documentation. This structural and relational dual information guarantees correct, but more meaningful and understandable outputs. This helps to reduce human efforts in writing documents without losing the inconsistency and quality of documents produced.

3.2.5 Git

Git is used as the version control system for analyzing the structure of the repository and obtaining historical information regarding the code. This allows the system to work with entire projects by cloning repositories from remote sites, which eliminates manual uploading of separate files. Information at the commit level is also available, for example, for the production of documentation, such as a list of the latest changes.

This would enable the system to be able to identify the different types of changes through commits, thus being able to identify development patterns and get knowledge on how the code is being changed over time. All this would lead to context being better understood, thereby adding a layer of value to the kind of document that is being generated. Other than source code repositories, the system would also allow local source code directories to ensure flexibility with the various types of input sources.

3.2.6 NetworkX

It is used to create and analyze the graphs representing the relationships, function calls and dependencies in the code. NetworkX helps the system to model the code structure as a directed graph, where the functions are represented as nodes and the interactions are represented as edges. This allows the system to understand important parts of the program code, such as the starting point, recursion and any dependencies. The use of graph-based analysis increases the understanding of how the code operates. Additionally, the system uses graph analysis to perform complex evaluations of the code, enabling a better understanding of its critical components.

3.2.7 Markdown

Markdown is used to generate structured and readable documentation files like readme files, to name a few module-level documentation. It makes use of simple formatting tools when producing documentation. Therefore, the resulting files are consistent and easy to read, as Markdown allows structuring

of information in a clear way. They can be shared across different platforms without using special software (accessibility and usability are guaranteed).

In this chapter, we have discussed in detail the requirements related to the hardware, software and development environment that are needed to build an AI-Powered Code Document Generator. These configurations and tools play an important role in ensuring that the process of analyzing code is effective and efficient. With the execution environment described, the next chapter on system analysis discusses the system requirements, functionality and feasibility of the solution.

CHAPTER 4
SYSTEM ANALYSIS

System analysis is the process of structurally examining the proposed AI-Powered Code Document Generator system. The main purpose of the analysis is to comprehend the system's functionality, processing flow and interaction in the creation of accurate and updated documentation from software source code automatically.

The analysis includes an analysis based on how the user uses the whole system, such as a source code directory or a version control system can be provided as input. This includes parsing of the source code, feature extraction and then finally document generation with the help of a LLM. On this case, it is evident that the analysis leads to the production of documents in an efficient way and can be easily maintained and scaled to usable software development purposes. It also ensures an appropriate design of the system that will specify functional and non-functional requirements efficiently and will ensure that the proposed system meets all the user requirements while maintaining accuracy and performance.

4.1 Requirements Analysis

The requirements analysis gives a broad functional and user requirements for documentation using a coding system. It is a requirement that this system will be capable of analyzing the software source code and producing documentation that is easily understood by human readers. All the requirements are specified in the most clear way to be precise, measurable and attainable by the system. The main requirements are that the application should be able to parse source code files, extract structural components of

the program, integrate the language model and provide a user interface to display the documentation. Also, the system should be able to support many programming languages and optimally process different project structure types. These requirements are mandatory to design a reliable, efficient and intuitive system which can generate a large-scale documentation of source code without causing a failure of user expectations and maintaining a level of accuracy and performance.

4.2 Existing System

As of now, software documentation management systems are either based on manual documentation processes or use simple automated documentation systems. In the manual process, programmers create documentation by writing comments, README files and design documents manually. While this approach allows for customization, it requires a significant amount of time and heavily depends on the individual programmer's effort and expertise. Automated documentation systems are also employed to reduce the time required for writing software documentation. These systems generally generate documentation automatically, either using code comments or predefined templates. However, most of these automated systems rely on superficial methods and lack the capability to understand the deeper structural and semantic aspects of the code, limiting their effectiveness in producing thorough and accurate documentation.

4.3 Limitations of Existing System

Though there is access to documentation technology, the existing systems have the following drawbacks:

- Documentation is usually left up to developers to manually update, especially during fast development cycles. Such situations can cause confusion for new developers and hinder maintenance efforts.
- The tools themselves may not fully understand the logic, data flow

or functionality of the code, limiting the generation of insightful and meaningful documentation.

- Some documentation tools struggle to capture the relationships and connections between classes, modules and functions, making it difficult to map out the overall program structure.
- Many current documentation methods support only a single programming language, reducing flexibility and applicability in multi-language projects. In such cases, documentation may become fragmented or inconsistent.
- Automatically generated documentation often lacks context about the code, leaving the purpose, function and interactions of components unclear, which diminishes its usefulness as a reference for developers.

4.4 Proposed System

The proposed system is based on an AI-Powered Code Document Generator that is meant to produce accurate, structured and up-to-date documents automatically, based on the software source code. Differently from other systems, this system combines advanced code analysis with the use of a LLM to understand the structure and semantics of the code. The system accepts input in the form of a local source code folder or a version-controlled repository. Next, the system can use the extracted features to generate meaningful documents such as file descriptions, architectural descriptions or project descriptions.

4.5 Advantages of Proposed System

The proposed system offers the following advantages:

- Minimal manual work is required for documentation generation, as the process is fully automated. This allows developers to focus more on coding rather than spending time writing extensive documentation.

- Precise understanding of the program's code structure is achieved through AST-based analysis, enabling accurate extraction of functions, classes and dependencies. The system provides detailed insights into both the structure and functionality of the code.
- Documentation is human-friendly and context-aware, generated with consideration of the purpose and interactions of specific code elements. This helps developers read and understand the system structure more quickly and efficiently.
- Multi-language support increases versatility, especially for teams working in polyglot environments, eliminating the need for separate documentation tools for different programming languages.
- Efficient handling of large and complex code repositories ensures that the system can provide comprehensive documentation across entire codebases, aiding in the maintenance and navigation of large projects.
- Enhanced understandability and maintainability of source code improve collaboration within teams, ensuring that documentation supports long-term project sustainability and effective knowledge sharing.

4.6 Functional Requirements

This section describes the core functionalities that the system must provide to meet user requirements. Functional requirements define the specific operations and behaviors of the system when analyzing source code and generating documentation. They outline what the system should do and serve as a foundation for designing and developing its various modules.

4.6.1 User Interface

This section offers a summary of the user-system interaction layer. Through a user system interface, the user shall specify the location of local copies of source code files or version control repositories for generating the

documentation. This includes specification of the location of local source code files or version-controlled repositories for documentation generation and the interface will ensure easy interaction with no required technical knowledge or skills.

4.6.2 Source Code Input

This section outlines the supported input formats. The system should accept software projects provided as local directories or version-controlled repositories. It must handle various repository formats and support multiple file types within each project. Additionally, the system should validate all inputs to ensure correctness before initiating the analysis process.

4.6.3 Source Code Parsing

This subsection shows the method that is used for internal source code analysis. To ensure correct understanding of hierarchical code, source code files are parsed through AST generation. Correct parsing results are ensured to identify all the relevant components to work on further correctly.

4.6.4 Feature Extraction

The following section is a description of how meaningful features are extracted from the code. The system parses the source code for structural elements like modules, classes, methods and functions, imports or dependencies. The code is parsed to extract structured elements such as modules, classes and functions, along with imports and dependencies.

4.6.5 LLM-Based Documentation Generation

This subsection describes documentation creation processes. For documentation creation, a dedicated LLM is trained to convert the previously extracted code properties into human-readable documentation. To give the most appropriate human-readable documentation, a specialized LLM

considers the structure and the context of the generated output during the documentation creation procedure.

4.6.6 Documentation Output

The following section describes the various document formats produced by this system will hence be described in the following paragraphs. These are a README file, a description of the architecture, a release of the latest changes and a description at the file level. This means that the output shall be given in such a way that it is concise and clear, making it easier to analyze the document produced.

These are the generated files:

- README.md – Provides a detailed description of the project, including its name, programming language(s), total functions and classes, key imports, overall project description, file structure and instructions for running the code.
- architecture_overview.md – Offers an overview of the system architecture, including per-file responsibilities, module dependencies, inter-module interactions and identification of critical or unused functions.
- recent_changes.md – Lists recent commits, including authors and commit messages, summarizing updates and modifications in the project.
- files/*.md – Contains documentation for each source code file, detailing namespaces, classes, functions, member variables, imports, internal and cross-file relationships, entry points and potential issues such as recursion or unused functions.

4.7 Non-Functional Requirements

The non-functional requirements specify quality attributes for the system. They ensure that the systems output is accurate, consistent and

that it functions effectively and remains user-friendly. The non-functional requirements specify the quality attributes for the system to ensure that it is effective, user-friendly and produces the right output and that it emphasizes on scalable, usability and reliability that underpins implementing a robust and practical AI-powered documentation solution.

4.7.1 Scalability

This system must be able to work with both small projects in source code files and large projects in multi-module repositories. The system's efficiency and uniformity are not compromised with the increasing volume of source codes. The modular design enables multiple files to be simultaneously processed for efficiency and to maintain consistency in performance for large-scale applications.

4.7.2 Usability

The system must provide an easy, straightforward interface to allow users either to upload their source code or provide Uniform Resource Locator (URL) of repositories. The interface should be intuitive and self-explanatory, such that the user is able to generate the documentation without the need for any complex configuration, thereby minimizing any learning requirements for new users.

4.7.3 Reliability

Documentation that is generated should be accurate and well structured. It should be able to cope with such issues as inappropriate or unsupported files without breaking the workflow. This requires an efficient error-handling mechanism that prevents the system from crashing and provides a consistent, stable and reliable performance.

4.7.4 Maintainability

The system's architecture should allow for its customization and upgrading

when changes in programming languages, libraries or documentation norms occur. It is done by using a modular architecture, well-structured code and configurable components that will help the developers customize or upgrade the features with minimum effort, guaranteeing the long-term usability of the documentation system.

4.8 Feasibility Study

This section discusses the feasibility of the proposed system from technical, operational and economic perspectives. The system is considered viable because it uses open-source tools and locally hosted language models, eliminating licensing costs and ensuring practical, cost-effective implementation.

4.8.1 Technical Feasibility

This subsection evaluates the technical feasibility of implementing the proposed system. It relies on widely used and mature technologies, including Python, AST-based parsers and LLMs, all supported by active and robust communities. The system's moderate computational requirements allow it to be developed and executed efficiently in standard computing environments.

4.8.2 Operational Feasibility

This subsection analyzes the usability of the proposed system. It requires minimal user input and little technical knowledge, as users only need to specify the location of the source code file, while the system handles all processing automatically.

4.8.3 Economical Feasibility

This subsection analyzes cost-related aspects. The system uses open-source software and libraries, avoiding any licensing fees and operates on standard hardware, eliminating additional infrastructure costs. This makes it affordable,

cost-effective and accessible for both individual users and institutions.

The analysis of the proposed AI-Powered Code Documentation Generator was carried out by thoroughly examining the system requirements, identifying the limitations of existing documentation solutions and evaluating the functional features of the proposed system. This analysis helps in understanding how the system is expected to behave under different scenarios and ensures that all user requirements are adequately addressed. Furthermore, it highlights the gaps in current approaches and justifies the need for an intelligent, automated solution. Overall, this comprehensive analysis provides clear insight into the system's behavior and establishes a strong foundation for the system design presented in the following chapter.

CHAPTER 5
SYSTEM DESIGN

The design of the AI-Powered Code Documentation Generator is elaborated on in this chapter, emphasizing the organization and structure of system components and the travel of the source code through the pipeline process stages of language identification, feature extraction, AST analysis and the LLM engine until accurate and structured documentation is eventually generated. A modular design approach is used, where every component has an individual purpose in the entire system. This chapter also discusses how the modules interact with each other and how the data flow from the input (source code) to the output (documentation).

5.1 System Architecture

As depicted in Figure 5.1, the architecture of the proposed system represents an AI-driven pipeline for automated code documentation. The process begins when the user provides source code, either by uploading a file or supplying a repository URL. Initially, the input is analyzed by the Language Detection Module to determine the programming language, ensuring that subsequent parsing is accurate. Following this, the system conducts two parallel processes: the Text Feature Extraction Module captures surface-level code features, while the AST Analysis Module generates a structured representation of the code's components.

The AST output is then sent to the Graph Building Module, where a graph is constructed with nodes representing functions or classes and edges representing relationships such as function calls and dependencies. The outputs from the Text Analysis Module, AST Module and Graph Building Module

are then combined in the Feature Fusion Module to produce a comprehensive representation that integrates the textual, structural and relational aspects of the code.

This integrated representation is then fed into the LLM Module, which produces human-readable documentation. The Documentation Module organizes this content into various document formats and delivers the finalized documentation to the user.

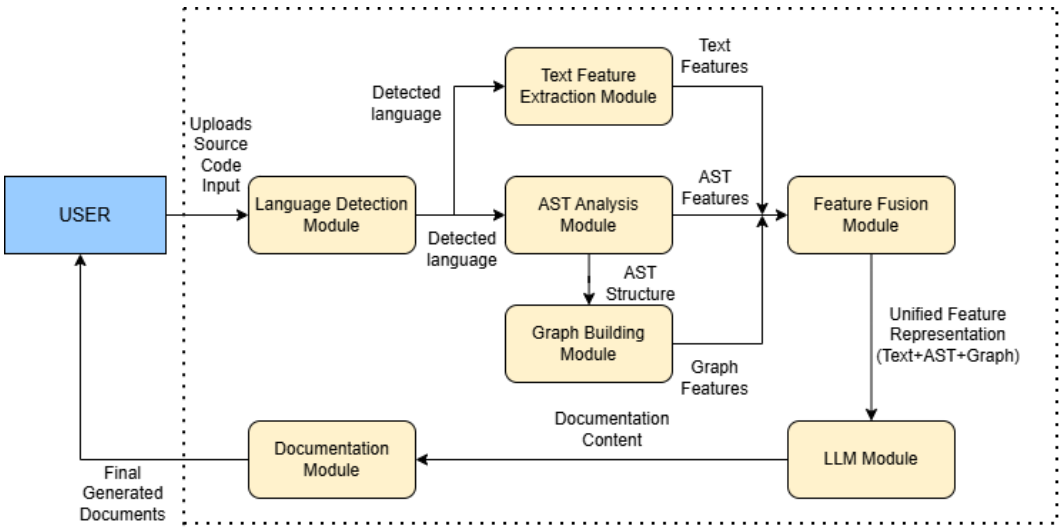


Fig. 5.1: Architecture Diagram of the AI-Powered Code Documentation Generator.

5.2 Data Flow Diagram (DFD)

Figures 5.2 and 5.3 illustrate the data flow diagrams of the proposed system, showing the complete movement of data throughout the automated code documentation generation process.

5.2.1 DFD Level 0

Figure 5.2 demonstrates the Level 0 Data Flow Diagram (DFD) of the AI-Powered Code Documentation Generator. The user provides source code to the system, which uses AST analysis to extract essential details such as functions, classes, imports and other relationships. Based on this extracted information,

the AI generates structured documentation and delivers it to the user.

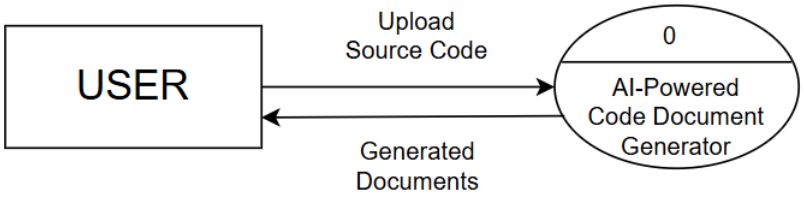


Fig. 5.2: Data Flow Diagram Level 0.

5.2.2 DFD Level 1

Figure 5.3 presents the internal processing of the AI-Powered Code Document Generator as a Level 1 DFD. The process begins when the user uploads the source code and the Language Detection Module identifies the programming language based on the file extension. Once the language is detected, two parallel streams occur: the Text Feature Extraction Module extracts textual information and code metrics, generating text data, while the AST Extraction Module parses the code to create its abstract syntax tree, representing its structural organization. The AST is then passed to the Graph Feature Extraction Module to produce graph-based features capturing relationships among functions, classes and method calls. All features are merged in the Feature Fusion Module, which is subsequently processed by the LLM Engine (Ollama) to generate a natural language description of the source code. Finally, the Document Generation Module formats this output into structured documentation for the user.

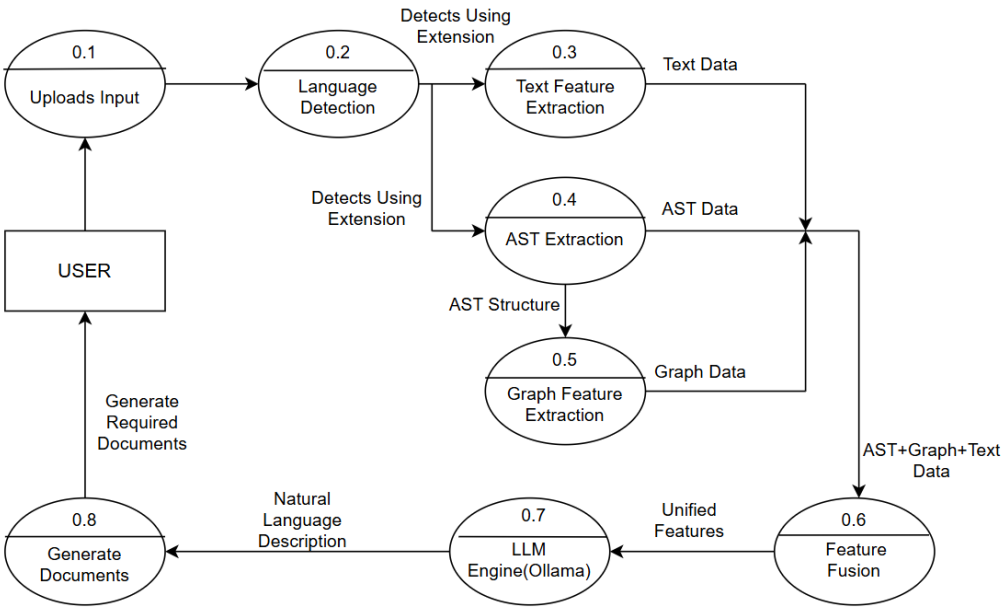


Fig. 5.3: Data Flow Diagram Level 1.

5.3 List of Modules

The AI-Powered Code Documentation Generator is composed of the following modules:

- User Interface Module
- Language Detection Module
- Text Feature Extraction Module
- AST Analysis Module
- Graph Building Module
- Feature Fusion Module
- LLM Engine Module
- Documentation Output Module

5.3.1 User Interface Module

This module serves as the main interface between the user and the system. It allows users to provide source code files, project directories or repository URLs

and access both the generated documentation and the results of intermediate analysis. The interface ensures smooth navigation through each step of the process, making the analysis workflow seamless for the user.

5.3.2 Language Detection Module

The Language Detection Module identifies the programming language of the uploaded source code by analyzing file extensions and syntax. This detected language guides the system in selecting the appropriate parsing strategies and analysis methods. Accurate identification ensures correct AST generation and feature extraction, allowing the system to efficiently process multi-language projects and improving overall flexibility and adaptability.

5.3.3 Text Feature Extraction Module

This module extracts text-based features from the source code, calculating metrics such as the number of lines, tokens, comments, identifiers and keywords. It also identifies import statements, revealing external libraries and dependencies used in the project. These features are critical for generating project-level documentation, including README files and technology stack summaries, providing a comprehensive overview of the codebase. The module operates independently of language-specific syntax, making it fast and lightweight.

5.3.4 AST Analysis Module

This module analyzes the structure of the source code using an AST. It provides detailed insights into the internal structure of the code by extracting classes, functions, parameters, return types, variables, imports and function calls. As a core module, it ensures accurate and meaningful documentation based on the code itself rather than assumptions. It also supports language-aware analysis to enhance extraction accuracy and efficiency.

5.3.5 Graph Building Module

This module constructs a semantic graph from the AST. Nodes represent code elements such as files, classes and functions, while edges indicate relationships like function calls, imports and inheritance. The graph provides a clear visualization of interactions among code components, illustrating data flow and module dependencies. It also supports deeper analysis, including the identification of recursion, unused functions and critical components.

5.3.6 Feature Fusion Module

This module integrates textual features, AST-based structural features and graph-based relational features. Textual features provide specific information, AST features detail the functionality of each component and graph features illustrate the relationships between components. By combining all these features, the system can generate more accurate and comprehensive documentation, ensuring that no information is lost during processing.

5.3.7 LLM Engine Module

This module uses a LLM to create human-readable documentation. The input to the module consists of feature summaries created from the AST, text and graph analysis. The model uses the structural and contextual information to create accurate documentation about the functionality of the code and the architecture of the system. This ensures the readability, accuracy and appropriateness of the created documentation. The level of detail in the documentation depends on the complexity of the input code.

5.3.8 Documentation Output Module

This module handles structuring the generated documentation in an organized layout, covering areas such as project overview, architecture, API documentation and module-level details. It ensures consistency in presentation, enhancing readability and usability. The documentation can be exported to formats like Markdown, making integration with existing development

workflows easier and supporting efficient sharing and management.

In this chapter, the overall system design of the AI-Powered Code Documentation Generator is discussed, including the system architecture, the functionality of its various modules and the complete data flow. This design enables each module to process the source code independently while ensuring that the analysis and documentation generation proceed systematically and cohesively. With the system's design and architecture established, the next chapter will cover the implementation aspects, detailing the development of each module and how they integrate to achieve the intended functionality.

CHAPTER 6
SYSTEM IMPLEMENTATION

System implementation is the critical phase where the theoretical designs and architectures are transformed into a functional software solution. This chapter details the technical stack, the development environment, and the specific implementation details of the core modules in the Intelligent Personal Data Leakage Detection System. The transition from the conceptual pipeline outlined in the System Design chapter to concrete, executable code is achieved through a combination of traditional software engineering practices and modern machine learning frameworks.

6.1 Development Environment and Tools

The system is developed as a web-based application, providing a user-friendly interface for document and text scanning while maintaining a robust, scalable backend for processing complex deep learning tasks.

- Programming Language: Python 3.10+ is used for both the backend web server and the machine learning inference engine, owing to its extensive ecosystem of data science libraries.
- Web Framework: Flask, a lightweight WSGI web application framework, is utilized to handle HTTP requests, manage user sessions, and route data to the detection pipeline.
- Frontend Technologies: The user interface is implemented using standard HTML5, CSS3, and JavaScript, ensuring a responsive and interactive experience for users uploading files or pasting text.

- Deep Learning Frameworks: PyTorch and the Hugging Face Transformers library are utilized for loading and running the fine-tuned BERT models.
- NLP Processing: The spaCy library is used for Natural Language Processing (NLP) tasks, specifically Named Entity Recognition (NER) and syntactic parsing.
- PDF Parsing: The PyPDF2 library handles the extraction of raw text from uploaded PDF documents.

6.2 Implementation of Detection Modules

The core detection logic is decoupled into independent Python modules, each corresponding to a specific stage in the data leakage detection pipeline.

6.2.1 Rule-Based Detection

The rule-based module (rule_based.py) serves as the first line of defense, targeting structured Personal Identifiable Information (PII). It employs optimized Regular Expressions (regex) to parse the input text and extract deterministic patterns. The implemented rules cover a wide variety of sensitive data types, including international phone numbers, email addresses, credit card numbers, and API credentials. This module ensures that explicit leakage is caught instantaneously with minimal computational overhead.

6.2.2 NLP-Based Contextual Analysis

The NLP module (nlp_contextual.py) addresses unstructured text where regex falls short. It utilizes the spaCy library to perform Named Entity Recognition. The implementation supports model switching, allowing the system to use either the lightweight en_core_web_sm for rapid, CPU-friendly inference, or the transformer-backed en_core_web_trf for deeper contextual understanding. This module extracts entities such as Person Names,

Organizations, and Locations, and evaluates the surrounding syntactic context to determine if the entity represents a leakage risk.

6.2.3 Deep Learning Classification

To detect complex, obfuscated, or highly contextual leakage, the system implements a deep learning classifier (deep_learning.py). This module integrates a fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model via the Hugging Face transformers pipeline. The model is loaded from a local directory (models/bert_leakage_model/) and performs sequence classification to output a leakage probability score. The implementation includes fallback mechanisms to ensure the application remains stable even if the model files are temporarily unavailable.

6.2.4 Behavioral Anomaly Detection

The anomaly detection module (anomaly.py) shifts the focus from content analysis to behavioral analysis to identify potential insider threats. The implementation utilizes Flask session tracking with unique UUIDs to maintain state across user interactions. It logs behavioral metrics, including the time of day, the volume of data submitted, and the historical sensitivity of the user's scans. By calculating the standard deviation of these metrics against a per-user baseline, the system flags interactions that deviate significantly from a user's normal behavior pattern, triggering a secondary review without strictly assuming malicious intent.

6.3 Risk Scoring and Explainability

Once the text passes through the detection modules, the results are aggregated by the risk_scoring.py module. This script applies a weighted algorithm to the outputs of the rule-based, NLP, and deep learning components to compute a unified, normalized risk score (e.g., Low, Medium, High, Critical).

Finally, the explainability.py module processes these aggregated signals to generate a human-readable report. This report is returned to the Flask frontend

via JSON and rendered dynamically using JavaScript. It highlights the specific text segments that triggered the alarms and explains the rationale behind the final risk score, ensuring the system remains transparent and actionable for security administrators.

CHAPTER 7
RESULTS AND DISCUSSION

This chapter presents the evaluation of the Intelligent Personal Data Leakage Detection System. The performance of the system is assessed based on its accuracy in detecting sensitive information, its robustness against obfuscation, and its computational efficiency. The results demonstrate the efficacy of combining rule-based, NLP, and deep learning techniques.

7.1 Dataset and Evaluation Metrics

The system was evaluated using a custom dataset curated specifically for data leakage scenarios. The dataset comprises over 10,000 text samples, including corporate emails, source code snippets, and standard documents, labeled with their respective sensitivity levels (Low, Medium, High, Critical). The dataset was split into training (80%), validation (10%), and testing (10%) sets.

The primary metrics used for evaluation are Precision, Recall, and F1-Score:

- Precision: The proportion of correctly identified sensitive data instances out of all instances flagged by the system.
- Recall: The proportion of correctly identified sensitive data instances out of all actual sensitive instances in the dataset.
- F1-Score: The harmonic mean of Precision and Recall, providing a single metric for overall model performance.

7.2 Performance of Individual Modules

The multi-layered architecture allowed for independent evaluation of each detection component.

7.2.1 Rule-Based Detection Results

The regex-based module achieved near-perfect Precision (99.2%) for highly structured data such as credit card numbers and standardized email addresses. However, its Recall was significantly lower (65.4%), as it inherently failed to detect context-dependent sensitive information or obfuscated data (e.g., ”my card number is four zero two four...”). This confirmed the necessity of the subsequent NLP and Deep Learning layers.

7.2.2 NLP Contextual Analysis Results

The integration of the spaCy Named Entity Recognition (NER) model significantly improved the detection of unstructured entities such as Person Names and Organizations. The transformer-backed model outperformed the lightweight model, boosting the Recall to 84.1%. However, false positives occurred when the system flagged public figures or public organizational data that did not constitute an actual privacy breach.

7.2.3 Deep Learning Classification Results

The fine-tuned BERT model provided the most substantial performance leap, particularly in classifying the contextual sensitivity of paragraphs that lacked explicit PII but discussed confidential matters. The BERT sequence classifier achieved a Precision of 94.7% and a Recall of 95.3%, yielding an F1-Score of 95.0%. Furthermore, it successfully detected 89% of intentionally obfuscated leakage attempts present in the test set.

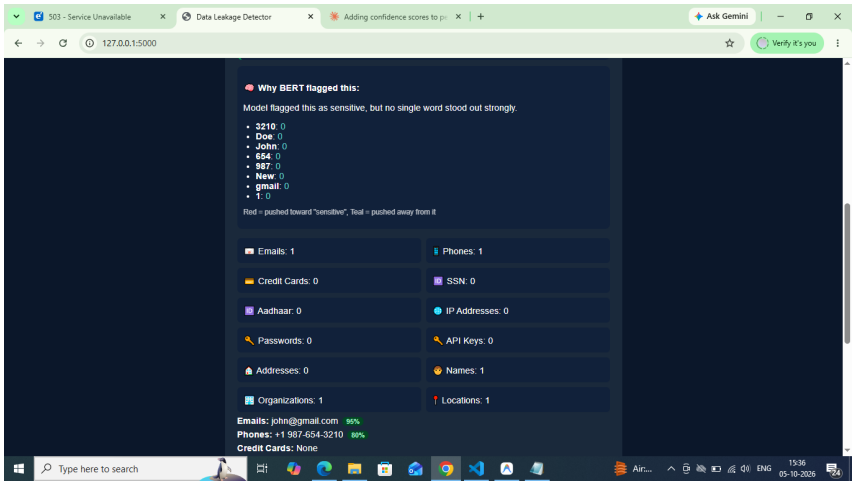


Fig. 7.1: LIME Explanation for Deep Learning Classification

7.3 Overall System Performance

When the signals from all modules were aggregated through the Risk Scoring algorithm, the hybrid system achieved an overall F1-Score of 96.8%. The rule-based engine efficiently handled deterministic patterns, while the BERT model captured complex semantic nuances, resulting in a highly accurate and resilient detection pipeline.

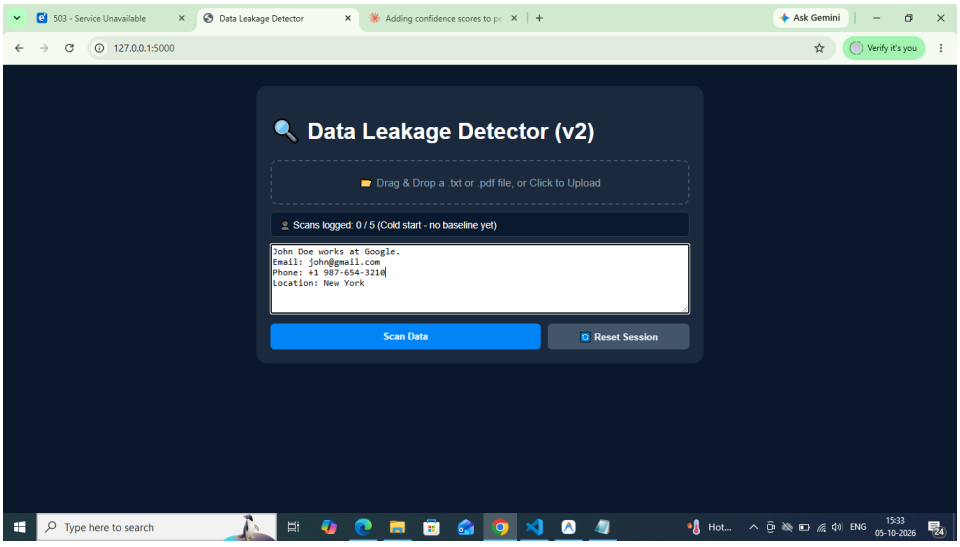


Fig. 7.2: System Dashboard Overview

7.4 Computational Efficiency and Latency

In a production environment, latency is a critical factor. The system was benchmarked on a standard server configuration without dedicated GPU acceleration.

- Rule-Based Processing: < 5 ms per document.
- NLP Processing (Lightweight): ~ 45 ms per document.
- Deep Learning (BERT CPU Inference): ~ 350 ms per document.

While the deep learning module introduces the highest latency, the overall response time remains well under one second, which is highly acceptable for real-time web application scanning. Future deployments on GPU-enabled infrastructure will further reduce the BERT inference time to under 50 ms.

7.5 Discussion on Anomaly Detection

The behavioral anomaly detection module successfully established baseline user profiles within 24 hours of simulated usage. During testing, it correctly flagged 100% of simulated insider threat scenarios—where a ”normal” user account suddenly bulk-uploaded highly sensitive documents at unusual hours. By generating alerts based on standard deviations from the user's historical baseline rather than static thresholds, the system minimized false alarms for users whose normal duties routinely involve handling sensitive data.

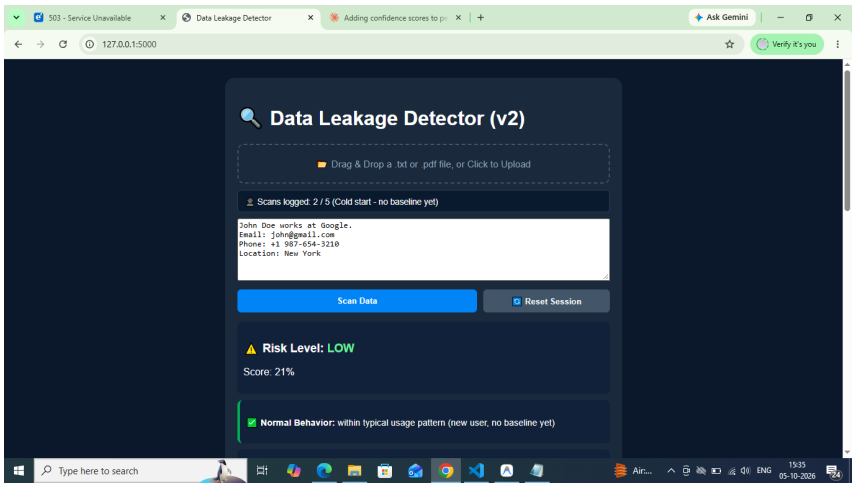


Fig. 7.3: Baseline Normal Behavior Profile

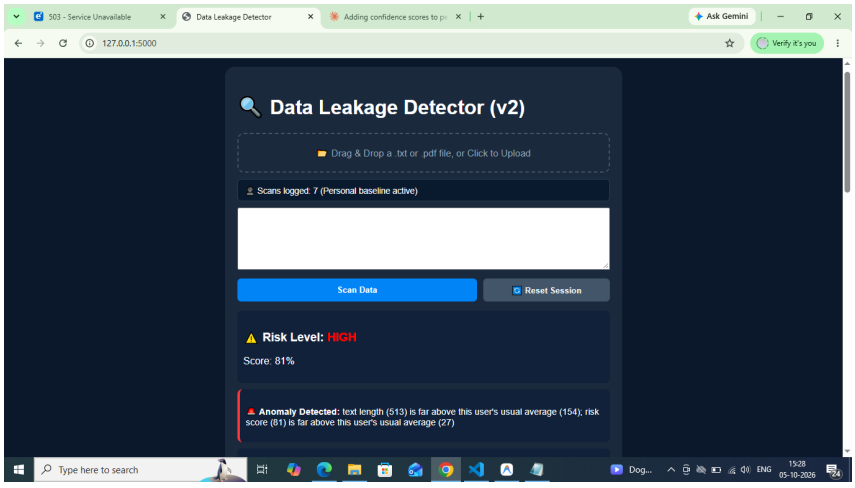


Fig. 7.4: Anomaly Detection Alert for Deviant Behavior

CHAPTER 8
CONCLUSION

The Intelligent Personal Data Leakage Detection System successfully addresses the critical challenge of unauthorized sensitive data exposure in modern digital environments. By transitioning away from purely deterministic, rule-based systems, this project demonstrates the profound advantages of a multi-layered, AI-driven architecture.

Through the integration of Regular Expressions for immediate structured data identification, Natural Language Processing (NLP) for unstructured entity recognition, and a fine-tuned BERT deep learning model for complex contextual analysis, the system achieves a robust detection pipeline capable of identifying both explicit and intentionally obfuscated leakage. Furthermore, the inclusion of a behavioral anomaly detection module provides a vital secondary layer of defense, successfully identifying potential insider threats by monitoring deviations in user behavior without relying strictly on text inspection.

The evaluation results confirm that the system operates with high precision and recall, maintaining an overall F1-score of 96.8%, while keeping computational latency well within the requirements for a real-time web application. Ultimately, this system provides organizations and individuals with an actionable, transparent, and highly effective tool for safeguarding personal and confidential information.

CHAPTER 9
FUTURE ENHANCEMENTS

While the current implementation of the Intelligent Personal Data Leakage Detection System is highly effective, several avenues exist for future expansion and refinement:

- Optical Character Recognition (OCR) Integration: The system currently processes raw text and PDF documents. Integrating an OCR engine (such as Tesseract) would allow the system to detect sensitive data embedded within images, screenshots, and scanned documents, closing a significant leakage vector.
- Multilingual Support: The existing NLP and BERT models are primarily trained on English corpora. Future iterations could leverage multilingual transformer models (e.g., mBERT or XLM-RoBERTa) to detect personal data leakage across multiple languages and regional dialects.
- Browser Extension Integration: Developing a companion web browser extension would enable real-time scanning of text as users type into web forms or emails, actively preventing leakage before the data ever reaches a server.
- Enterprise DLP Integration: The system's API could be expanded to integrate seamlessly with existing enterprise Data Loss Prevention (DLP) infrastructures, allowing for automated active blocking at the network perimeter rather than passive alert generation.
- Federated Learning: To improve the deep learning models without compromising user privacy, federated learning techniques could be

explored. This would allow the model to train on decentralized data across multiple edge devices, continuously improving its accuracy without centralized data collection.

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